College of Engineering and Technology

Project Work – Student Hand Book

B-1012

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Title of the Project:	CROPSIGHT: - An Explainable Risk Scoring Framework for Post Harvest Supply Chains
Project Site / Location	SRMIST, Kattankulathur
Name and address of the company / organisation (Applicable for projects with industry or industry support)	SRMIST, Kattankulathur, Chengalpattu District-603203

Supervision Team

	Supervisor	**Co-Supervisor**	**External Supervisor (If applicable)**
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Degree/ program	B.Tech	**Specialisation**	Computer Science & Engineering

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Academic Year	2025-2026 (Even)	**Semester**	6
Course Code	**21CSP302L**	**Course Title**	Project

Mission Statement

To design and develop a machine learning based predictive risk modeling framework that estimates post-harvest waste risk in Indian agricultural supply chains using multi-source proxy indicators, enabling early risk identification, explainable decision support, and sustainable interventions to reduce food loss.

Problem (or) Product Description:

India faces significant post-harvest losses due to unfavorable storage conditions, transportation delays, climatic stress, and market inefficiencies. A major challenge is the absence of unified, structured datasets and predictive systems that can estimate the likelihood of waste before actual spoilage occurs. Most current approaches are reactive rather than preventive.

This project proposes a **Machine Learning Based Predictive Risk Engine** that estimates the level of post-harvest waste risk using agricultural proxy indicators collected from multi-source datasets. The system preprocesses and integrates relevant factors, trains classification models, and predicts the waste risk category of produce as **Low, Medium, or High**. The best-performing model is further extended into a probabilistic **risk scoring engine** that generates a numerical risk score and assigns a risk zone.

The framework also includes explainability mechanisms to identify the factors most responsible for elevated waste risk, making the system useful not only for prediction but also for decision support. The project is intended as a research-oriented prototype for data-scarce agricultural environments and can be extended in the major project phase using real-time datasets, IoT sensors, and live market information

Assumptions and Constraints

Assumptions

1. Due to the lack of unified Indian post-harvest loss datasets, the study uses **multi-source agricultural proxy data** to approximate waste risk conditions.
2. The selected features reasonably represent factors that influence post-harvest deterioration, such as environmental stress, crop-related conditions, and logistical or market-related indicators.
3. The generated target label **waste_risk** is treated as a valid approximation of actual post-harvest vulnerability for the purpose of model training.
4. The trained model can support **risk estimation** and prioritization, even though it is not yet validated in a live production supply chain environment.
5. The framework is intended as a **decision-support and research prototype**, not as a final industrial deployment system.

Constraints

1. No direct real-time post-harvest loss dataset is available in a consolidated Indian context.
2. The project timeline limits large-scale primary data collection and field validation.
3. The model relies on proxy indicators rather than directly measured spoilage outcomes.
4. Medium-risk cases may have overlap with low-risk and high-risk conditions, affecting perfect class separation.

Real-time deployment, IoT integration, and live API-based monitoring are outside the scope of the minor project and are reserved for future extension.

Stakeholders

1. **Farmers** – to identify produce batches that may be vulnerable to waste and take preventive action.
2. **Storage and Warehouse Operators** – to prioritize storage management for high-risk produce.
3. **Transport and Supply Chain Managers** – to identify and manage delay-sensitive agricultural movement.
4. **Agricultural Policy Makers and Government Agencies** – to understand risk patterns and support food-loss reduction strategies.
5. **Researchers and Academic Institutions** – to study predictive risk modeling in agriculture and extend the framework further.
6. **Wholesale Markets / Mandi Ecosystem Participants** – to estimate operational conditions contributing to delayed selling and wastage.
7. **Consumers and Society at Large** – as reduced post-harvest loss improves food availability, sustainability, and economic efficiency.

Division of work and contributors of SPRINT 1 [Include Daily Scrum of Sprint 1]

Field	Details
Sprint / Review	**January 2026 (SPRINT Review 1)**
Sprint Focus	**Dataset collection, preprocessing, target definition, and initial model development**
Project Title	**Machine Learning Based Predictive Risk Engine for Reducing Post-Harvest Food Waste in Indian Agriculture Supply Chains**
Team Members	**Vaidehi Mishra, Rishika Dhanuka**
Objective of Sprint 1	To establish the project foundation by identifying the problem statement, collecting relevant multi-source agricultural proxy datasets, cleaning and preprocessing the data, defining the target variable **waste_risk**, and training initial machine learning models for waste risk classification.
Work Done in Sprint 1	Conducted problem analysis and SDG mapping under Agriculture. Studied post-harvest waste challenges in Indian supply chains. Collected and examined proxy agricultural datasets. Performed exploratory data analysis, missing value handling, categorical encoding, and removal of irrelevant/high-cardinality attributes. Defined the target variable **waste_risk**. Froze the cleaned dataset and carried out train-test split. Trained Logistic Regression, Random Forest, and Gradient Boosting models. Evaluated performance using accuracy, recall, confusion matrix, and ROC-AUC. Selected Gradient Boosting as the best-performing model and converted predictions into a risk scoring framework with zones such as Low, Moderate, High, and Critical.
Contribution – Vaidehi Mishra	Led project ideation, problem formulation, SDG alignment, dataset collection, exploratory data analysis, preprocessing, encoding, model training, model evaluation, and risk engine formulation.
Contribution – Rishika Dhanuka	Contributed to literature review, dataset understanding, proxy variable mapping, validation of target formulation, preprocessing support, result

	interpretation, documentation, and framing of assumptions, constraints, and stakeholders.
Sprint 1 Outcome	A cleaned and finalized dataset of **50,000 records** with **163 encoded features** was prepared. Waste risk classes were defined as **Low, Medium, and High**. Among the tested models, **Gradient Boosting** performed best with approximately **79% accuracy** and **0.876 ROC-AUC**, forming the base of the predictive risk engine.
Review Status	**Completed**

Day	**Progress**
Day 1	Selected Agriculture as the SDG domain and finalized the project topic related to post-harvest waste reduction.
Day 2	Conducted background study on Indian agriculture supply chains and post-harvest loss challenges.
Day 3	Identified and collected relevant multi-source agricultural proxy datasets.
Day 4	Performed exploratory data analysis and studied feature distributions and dataset structure.
Day 5	Cleaned the data, handled missing values, and removed irrelevant or high-cardinality columns.
Day 6	Encoded categorical variables and prepared the final feature matrix for modeling.
Day 7	Defined the target variable **waste_risk** and checked class distribution.
Day 8	Performed train-test split and trained Logistic Regression, Random Forest, and Gradient Boosting models.
Day 9	Compared model performance using accuracy, recall, confusion matrix, and ROC-AUC.
Day 10	Selected Gradient Boosting as the final model and converted predictions into a preliminary risk scoring system.

Signature of the Supervisor

Division of work and contributors of SPRINT 2

Field	**Details**
Sprint / Review	**March 2026 (SPRINT Review 2)**

Sprint Focus	**Model optimization and explainability**
Project Title	**Machine Learning Based Predictive Risk Engine for Reducing Post-Harvest Food Waste in Indian Agriculture Supply Chains**
Team Members	**Vaidehi Mishra, Rishika Dhanuka**
Objective of Sprint 2	To improve the predictive performance and interpretability of the waste risk model by optimizing the selected machine learning algorithm, evaluating detailed performance metrics, converting predictions into a risk scoring engine, and incorporating explainability for identifying the major drivers of post-harvest waste risk.
Work Done in Sprint 2	Fine-tuned and finalized the Gradient Boosting model as the best-performing classifier. Evaluated the model using accuracy, class-wise recall, confusion matrix, and ROC-AUC. Observed that the model performed strongly for Low and High risk classes while Medium risk remained comparatively harder to separate. Converted class probabilities into a **Risk Score** ranging from 0–100 and mapped the score into four operational zones: **Low, Moderate, High, and Critical**. Verified that the **Critical** zone strongly aligned with actual High-risk cases, showing high practical reliability. Added explainability analysis using feature importance and model interpretation to identify the most influential variables contributing to waste risk. Reframed the system as a **proxy-based predictive risk modeling framework** suitable for data-scarce Indian agricultural settings.
Contribution – Vaidehi Mishra	Led model refinement, evaluation, optimization decisions, probabilistic risk score generation, risk zoning logic, interpretation of Critical risk bucket performance, and framing of the explainable predictive risk engine.
Contribution – Rishika Dhanuka	Assisted in performance analysis, documentation of model behavior, interpretation of explainability results, preparation of assumptions and limitations, and support in structuring the framework as a research-oriented decision-support system.
Sprint 2 Outcome	The optimized **Gradient Boosting** model achieved around **79% accuracy** with **0.876 ROC-AUC**. Recall values were approximately **0.89 for Low**, **0.75 for High**, and **0.61 for Medium** risk classes. The generated **Critical Risk Zone** contained **1003 samples**, of which **857 cases (~85.4%)** correctly corresponded

	to actual High-risk instances, demonstrating strong reliability of the risk prioritization framework.
Review Status	**Completed**

Daily Scrum of Sprint 2

Day	**Progress**
Day 1	Revisited Sprint 1 results and confirmed Gradient Boosting as the final model for optimization and extension.
Day 2	Re-trained the model on the finalized dataset and checked performance consistency on train-test split.
Day 3	Evaluated detailed classification metrics including accuracy, class-wise recall, confusion matrix, and ROC-AUC.
Day 4	Analyzed class performance and observed that Medium-risk class had comparatively lower recall than Low and High classes.
Day 5	Converted prediction probabilities into a quantitative **Risk Score (0–100)** to make the system more actionable.
Day 6	Designed and assigned risk zones: **Low, Moderate, High, and Critical** based on score thresholds.
Day 7	Validated the Critical risk bucket and confirmed that most Critical predictions aligned with actual High-risk samples.
Day 8	Performed explainability analysis to identify the most influential features driving waste risk predictions.
Day 9	Documented model limitations, especially the use of proxy agricultural indicators instead of direct post-harvest loss measurements.
Day 10	Consolidated Sprint 2 findings and framed the system as an explainable **proxy-based predictive risk engine** for Indian agricultural supply chains.

Signature of the Supervisor

Division of work and contributors of SPRINT 3

Field	Details
Sprint / Review	**April 2026 (SPRINT Review 3)**
Sprint Focus	**Final system integration and paper submission**
Project Title	**Machine Learning Based Predictive Risk Engine for Reducing Post-Harvest Food Waste in Indian Agriculture Supply Chains**
Team Members	**Vaidehi Mishra, Rishika Dhanuka**
Objective of Sprint 3	To consolidate the complete predictive risk framework, finalize the explainable waste risk estimation system, prepare all results and documentation, and submit the research paper based on the developed model and findings.
Work Done in Sprint 3	Finalized the complete machine learning based predictive risk engine by integrating the cleaned dataset, optimized Gradient Boosting model, probabilistic risk scoring logic, risk zone classification, and explainability analysis into one unified framework. Structured the final system as a **proxy-based predictive risk modeling framework** for data-scarce Indian agricultural environments. Prepared final result tables, model performance summaries, confusion matrix interpretation, ROC-AUC findings, and Critical zone validation. Documented assumptions, constraints, stakeholder impact, limitations, and future scope. Created the final architecture of the system showing data collection, preprocessing, model prediction, risk scoring, zoning, and decision-support outputs. Drafted, refined, and finalized the research paper for submission based on the project methodology, experiments, results, and conclusions.
Contribution – Vaidehi Mishra	Led final system consolidation, risk engine integration, architecture design, interpretation of final model results, framework documentation, and preparation of the research paper manuscript.

Contribution – Rishika Dhanuka	Assisted in final result validation, documentation of limitations and future scope, organization of paper sections, formatting of manuscript content, and support in submission readiness.
Sprint 3 Outcome	A complete and explainable **Machine Learning Based Predictive Risk Engine** was finalized for post-harvest waste risk estimation. The project was documented as a research-oriented framework and prepared for academic paper submission. The final system demonstrated strong identification of high-risk and critical cases and provided a solid foundation for extension into the major project phase.
Review Status	**Completed**

Daily Scrum of Sprint 3

Day	**Progress**
Day 1	Reviewed Sprint 2 outputs and listed all final components required for project closure and paper submission.
Day 2	Consolidated dataset preprocessing, trained model, and evaluation outputs into a final unified workflow.
Day 3	Finalized the probabilistic risk scoring logic and verified risk zone assignment consistency.
Day 4	Validated final model results including accuracy, recall, ROC-AUC, and Critical zone reliability.
Day 5	Structured the complete project as a proxy-based predictive risk modeling framework for Indian agriculture supply chains.
Day 6	Prepared final architecture diagram, workflow explanation, and system-level documentation.
Day 7	Documented assumptions, constraints, stakeholders, limitations, and future extension possibilities.
Day 8	Drafted the research paper sections including introduction, methodology, results, discussion, and conclusion.

Day 9	Refined the paper content, aligned project findings with SDG relevance, and finalized formatting for submission.
Day 10	Completed the final project documentation and submitted the paper/manuscript for academic review.

Signature of the Supervisor

Worksheet / Data collection / Observation etc

Section	Details
Project Title	**Machine Learning Based Predictive Risk Engine for Reducing Post-Harvest Food Waste in Indian Agriculture Supply Chains**
Type of Data Used	Multi-source agricultural proxy dataset
Source of Data	Publicly available datasets collected from Kaggle and combined for research purposes
Nature of Data	Secondary data
Total Records Collected	**50,000**
Final Number of Features	**163 features** after preprocessing and encoding
Target Variable	**waste_risk**
Type of Problem	Multiclass classification
Risk Classes	**Low, Medium, High**
Reason for Using Proxy Data	A unified real-world Indian post-harvest loss dataset with structured supply chain factors was not readily available within the project timeline. Therefore, multi-source agricultural proxy indicators were used to approximate waste risk.
Important Variables Considered	Crop type, environmental conditions, production-related indicators, storage-related proxy factors, and market-related proxy factors
Data Preprocessing Performed	Missing value handling, irrelevant column removal, high-cardinality attribute removal, categorical encoding, feature preparation, and dataset freezing for final modeling
Dropped Attribute	**District_Name** was removed due to very high cardinality (**642 unique values**)
Class Distribution Observed	Low – **26,132**, High – **12,500**, Medium – **11,368**

Class Balance Observation	No severe imbalance observed; largest class was around **52%**, so resampling was not required
Model Observation	Logistic Regression, Random Forest, and Gradient Boosting were trained and compared
Best Model Identified	**Gradient Boosting Classifier**
Final Performance Observation	Accuracy – **0.79**, ROC-AUC – **0.876**, High Recall – **0.75**, Medium Recall – **0.61**, Low Recall – **0.89**
Risk Engine Observation	Model probabilities were converted into a **Risk Score (0–100)** and mapped into **Low, Moderate, High, and Critical** risk zones
Critical Zone Observation	Out of **1003 Critical** cases, **857 cases (~85.4%)** correctly matched actual High-risk samples
Inference from Observation	The model showed strong ability to prioritize truly high-risk produce batches, making it suitable as a proxy-based predictive risk estimation framework
Limitation Observed	Medium-risk class showed lower recall than Low and High classes, indicating overlap between moderate and neighboring risk conditions
Outcome of Data Collection and Observation	The collected proxy dataset was sufficient to build, train, and evaluate an explainable waste risk prediction framework for Indian agricultural supply chain conditions

Research Article with Journal Publication Details / Patent disclosure form with patent status

Field	Details
Title of Research Article	**A Proxy-Based Machine Learning Framework for Predictive Post-Harvest Waste Risk Estimation in Indian Agricultural Supply Chains**
Authors	**Vaidehi Mishra, Rishika Dhanuka**
Type of Publication	Research Article
Domain	Machine Learning, Agriculture, Sustainable Development
Related SDG	**SDG 2 – Zero Hunger, SDG 12 – Responsible Consumption and Production**
Problem Addressed	Estimation of post-harvest waste risk in Indian agricultural supply chains using multi-source proxy indicators in a data-scarce environment
Abstract / Summary	This research proposes a machine learning based predictive risk modeling framework to estimate post-harvest waste risk in Indian agricultural supply chains. Due to the lack of unified real-world post-harvest loss datasets, the study uses multi-source agricultural proxy indicators to approximate waste vulnerability conditions. Multiple classification models were evaluated, and Gradient Boosting was selected as the best-performing model. The model was further extended into a probabilistic risk engine that generates numerical risk scores and assigns operational risk zones such as Low, Moderate, High, and Critical. The framework also supports

	explainability and decision support, making it suitable as a research-oriented prototype for data-scarce agricultural systems.
Status of Article	**Prepared / Submitted for publication** *(write whichever is true in your case)*
Target Journal / Conference	**Student conference / journal / institutional publication platform** *(replace with actual journal or conference name if finalized)*
Date of Submission	**April 2026**
Paper Outcome	Under Review / Submitted *(choose appropriately)*
Novel Contribution	Introduces a **proxy-based predictive risk modeling framework** for post-harvest waste estimation in Indian agriculture where direct structured datasets are unavailable
Method Used	Data preprocessing, feature engineering, multiclass classification, Gradient Boosting, probabilistic risk scoring, risk zoning, and explainability
Key Result	Final model achieved approximately **79% accuracy** and **0.876 ROC-AUC**, with the Critical risk zone showing around **85.4% alignment** with actual High-risk cases
Future Scope Mentioned in Paper	Real-time data integration, IoT-based monitoring, live market linkage, and extension into a deployable decision-support platform

Item	Details
Research Article Title	A Proxy-Based Machine Learning Framework for Predictive Post-Harvest Waste Risk Estimation in Indian Agricultural Supply Chains
Authors	Vaidehi Mishra, Rishika Dhanuka
Status	Prepared / Submitted
Submission Date	April 2026
Key Contribution	Proxy-based ML risk estimation framework for Indian post-harvest waste prediction
Best Model	Gradient Boosting
Key Performance	Accuracy: 0.79, ROC-AUC: 0.876
Proof Attached	Manuscript copy and submission acknowledgement / draft proof